

Effect of Doping on Semiconductor Transport Properties

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Introduction:

This study aims to explore the concept of doping in semiconductors and its impact on their electrical behavior. Doping involves the intentional introduction of impurity atoms into a pure semiconductor to modify its carrier concentration and enhance its conductivity. By controlling the type and level of doping, significant changes occur in key transport properties such as resistivity, conductivity, and carrier mobility. Understanding these effects is essential for analyzing and designing semiconductor devices with desired electrical performance.

Theoretical background:

1-Overview of Semiconductor Electrical Behavior

A semiconductor is a material whose electrical conductivity falls between that of a good conductor (like copper) and a good insulator (like rubber). At first glance, that sounds like a limitation. In reality, it is a superpower.

Unlike metals, whose conductivity is fixed, the electrical behaviour of a semiconductor can be modified by changing temperature, adding impurities, or applying electric fields. This tunability is what allows engineers to build transistors, diodes, LEDs, solar cells, and integrated circuits – all from the same class of materials.

Silicon is the most widely used semiconductor globally, accounting for the vast majority of transistors and microchips in production. Germanium, gallium arsenide (GaAs), and gallium nitride (GaN) are other important semiconductor materials, each with distinct electrical characteristics suited to specific applications.

Electrical properties of semiconductor materials determine how a device performs, how much power it consumes, and how reliably it operates over time. Understanding these properties is foundational to semiconductor material and device characterization.

2-Energy Band Theory of Semiconductors:

In any solid material, electrons occupy energy levels grouped into bands. The valence band contains electrons involved in chemical bonding, while the conduction band is where electrons move freely and contribute to electrical current. The energy gap between these two bands is called the band gap.

In conductors like copper, bands overlap, allowing electrons to flow freely with almost no energy input. In insulators like glass, the band gap is very large (above 5 eV), so electrons

simply cannot jump across. In semiconductors, the band gap is moderate, typically between 0.5 and 3 eV, which means electrons can be excited into the conduction band by thermal energy, light, or an applied electric field.

This is why silicon (with a band gap of about 1.1 eV) works effectively for electronic devices. Enough electrons cross the gap at room temperature to enable useful conduction, but the material is not so conductive that it cannot be controlled or switched off.

Band gap also determines optical and electronic behavior. Direct band gap materials like GaAs are used in LEDs and laser diodes because electron transitions between conduction and valence bands can directly produce photon emission.

3-Conduction in Intrinsic Semiconductors:

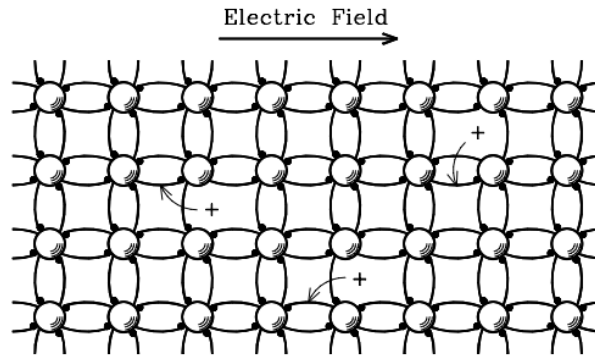
The thermal energy stored in a semiconductor crystal lattice causes the atoms to be in constant mechanical vibration. At room temperature, the vibrations shake loose several valence electrons, which then become free electrons. In intrinsic silicon, the number of free electrons is approximately one in 10^{12} of the total number of valence electrons. The free electrons behave similarly to those in a metal. Under the influence of an applied electric field, they have a mobility and exhibit a drift velocity which produces a conduction current. However, because of the small number of free electrons, the conductivity of an intrinsic semiconductor is much lower than that of a metal.

When an electron is shaken loose from an atom, an electron vacancy is left which is called a hole. The parent atom then becomes an ion. The constant mechanical vibration of the lattice can cause the ion to capture a valence electron from a neighboring atom to replace the missing one. When such a transfer takes place, the position of the hole moves from one atom to another. This is equivalent to a positive charge $+q$ moving about in the semiconductor. Like free electrons, holes have a mobility and exhibit a drift velocity that produces a conduction current under the influence of an applied electric field. Because of the opposite charge polarity of electrons and holes, they drift in opposite directions under the influence of a field.

In effect, a hole drifts to the right when a bound valence electron shifts to the left from one atom to another. In summary, the flow of current in the semiconductor is the result of the flow of two components. One component is the flow of free electrons in one direction. The other component is the flow of the absence of bound electrons in the other direction. Because of the opposite charge polarities, the electron current and the hole current add to produce the total conduction current.

In this case we write $n = p = n_i$, where n_i is called the intrinsic concentration. It can be shown that n_i can be written

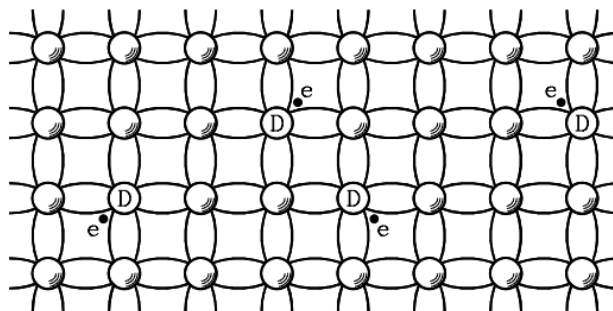
$$\bar{n}_i = n_0 T^{3/2} \exp\left(\frac{-V_G}{2V_T}\right)$$



4-n-Type Semiconductor:

An n-type semiconductor is produced by adding a donor impurity such as arsenic, antimony, or phosphorus to an intrinsic semiconductor. Each donor atom has five valence electrons. When a donor atom replaces an atom in the crystal lattice, only four valence electrons are shared with the surrounding atoms. The fifth valence electron becomes a free electron as illustrated in the Figure.

The number of free electrons donated by the donor atoms is much greater than the number of free electrons and holes in the intrinsic semiconductor. This makes the conductivity of the n-type semiconductor much greater than that of the intrinsic semiconductor. Because the number of free electrons is far greater than the number of holes, the free electrons are the majority carriers. The semiconductor is called n-type because the majority carriers have a negative charge.

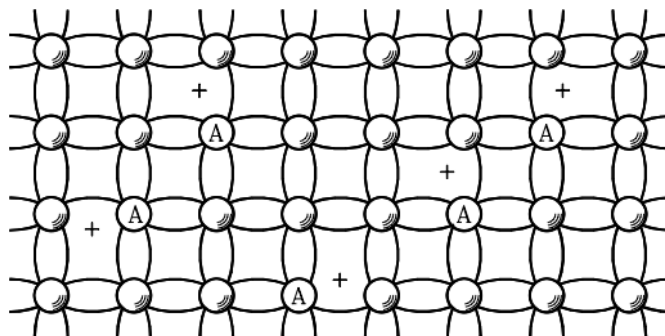


Because of the large number of donor electrons, there are many more free electrons available for recombination with the holes. This decreases the mean lifetime for the holes, which decreases the number of holes in the n-type semiconductor compared to the intrinsic semiconductor. For this reason, the current due to the flow of holes in an n-type semiconductor is often neglected in calculations.

5-p-Type Semiconductor:

A p-type semiconductor is produced by adding an acceptor impurity such as gallium, boron, or indium to an intrinsic semiconductor. Each acceptor atom has three valence electrons. When an acceptor atom replaces an atom in the crystal lattice, there are only three valence electrons shared with the surrounding atoms. This leaves a hole as illustrated in the Figure. The number of holes created by the acceptor atoms is much greater than the number of free electrons and holes in the intrinsic semiconductor. This makes the conductivity of the p-type semiconductor much greater than that of the intrinsic semiconductor. Because the number of holes is far greater than the number of electrons, the holes are the majority carriers. The semiconductor is called p-type because the majority carriers have a positive charge.

Hole-electron pairs are continually formed by thermal agitation of the lattice in a p-type semiconductor. Because of the large number of holes, there are many more holes available for recombination with the free electrons. This decreases the mean lifetime for the free electrons which decreases the number of electrons in the p-type semiconductor compared to the intrinsic semiconductor. For this reason, the current due to the flow of free electrons in a p-type semiconductor is often neglected in calculations.



In an intrinsic semiconductor, we have noted that the electron concentration and the hole concentration are both equal to the intrinsic concentration, i.e. $n = p = n_i$. If this were not

true, the material would not be electrically neutral. We have seen that adding an n-type impurity to the semiconductor increases n and decreases p. Similarly, adding a p-type impurity increases p and decreases n. It can be shown that the product of n times p is a constant independent of the doping type and the doping level. The product is given by :

$$np = n_i^2$$

6-Mobility and conductivity:

In solid-state physics, the electron mobility characterizes how quickly an electron can move through a metal or semiconductor when pushed or pulled by an electric field. There is an analogous quantity for holes, called hole mobility. The term carrier mobility refers in general to both electron and hole mobility.

Electron and hole mobility are special cases of electrical mobility of charged particles in a fluid under an applied electric field.

When an electric field E is applied across a piece of material, the electrons respond by moving with an average velocity called the drift velocity, v_d μ is defined as

$$v_d = \mu E.$$

Conductivity is proportional to the product of mobility and carrier concentration. For example, the same conductivity could come from a small number of electrons with high mobility for each, or a large number of electrons with a small mobility for each. For semiconductors, the behavior of transistors and other devices can be very different depending on whether there are many electrons with low mobility or few electrons with high mobility.

Therefore mobility is a very important parameter for semiconductor materials. Almost always, higher mobility leads to better device performance, with other things equal.

Semiconductor mobility depends on the impurity concentrations (including donor and acceptor concentrations), defect concentration, temperature, and electron and hole concentrations. It also depends on the electric field, particularly at high fields when velocity saturation occurs.

When an electric field $\vec{E}$ is applied to an intrinsic semiconductor, the free electrons drift with a velocity $\vec{v}_e = -\mu_e \vec{E}$, where μ_e is the electron mobility. The holes drift with a velocity $\vec{v}_h = +\mu_h \vec{E}$, where μ_h is the hole mobility.

Although the free electrons and holes drift in opposite directions, the current densities add because the charge polarities are opposite. The electron charge density is $\rho_e = -n_i q$ and the hole charge density is $\rho_h = +n_i q$, where n_i is the intrinsic concentration. The total conduction current density can be written

$$\vec{J} = \rho_e \vec{v}_e + \rho_h \vec{v}_h = n_i(\mu_e + \mu_h)q\vec{E} = \sigma\vec{E}$$

This equation defines the conductivity σ of the intrinsic semiconductor. It is given by

$$\sigma = n_i(\mu_e + \mu_h)q$$

So, Conductivity is proportional to the product of mobility and carrier concentration

Resistivity is inversely proportional to the conductivity; it is the reciprocal of it. The more conductivity is, the more current will pass through this material.

Effect of Doping on Mobility and Conductivity:

Dopants in an intrinsic semiconductor perform two major functions

- They increase carrier concentration of a particular polarity (i.e. electrons or holes) so that the overall conductivity is higher. Usually this is orders of magnitude higher than the intrinsic semiconductor and is dominated by either electrons or holes i.e. the majority carriers.
- Dopants also stabilize carrier concentration around room temperature. For Si, the saturation regime extends from roughly 60 K to 560 K.

Consider the general conductivity equation

$$\sigma = ne\mu_e + pe\mu_h$$

For extrinsic semiconductors with either n or p much higher than the minority carrier concentration equation 1 can be written as

$$\begin{aligned} n \text{ type : } \sigma &= ne\mu_e \\ p \text{ type : } \sigma &= pe\mu_h \end{aligned}$$

μ_e and μ_h are the drift mobilities of the carriers. They are given by

$$\begin{aligned} \mu_e &= \frac{e\tau_e}{m_e^*} \\ \mu_h &= \frac{e\tau_h}{m_h^*} \end{aligned}$$

The drift mobilities are a function of temperature and in extrinsic semiconductors they depend on the dopant concentration. This dependence comes from the scattering time, τ_e and τ_h , which is the time between two scattering events.

Another way of writing the scattering time is in terms of a scattering cross section (S). This is given by

$$\tau = \frac{1}{S v_{th} N_s}$$

where v_{th} is the mean speed of the electrons (thermal velocity) and N_s is the number of scatters per unit volume. The role of scatters and the effect on mobility can be considered by considering the two main types of scatters:

1-Lattice atoms

2-Impurities - dopants

Lattice scattering:

In an intrinsic semiconductor the carrier scattering is mainly due to thermal vibrations of the lattice (Si atoms). In metals, electron-electron scattering is also important and affects the conductivity, but electron concentration in semiconductors are low so that electron-electron scattering can be ignored. For lattice scattering, if a is the amplitude of the lattice vibrations for a simple two dimensional scattering, S is just πa^2 . As temperature increases, a increases and hence S increases. The second term in equation 4 is the thermal velocity (v_{th}). This is typically given by $\frac{3}{2} k_B T$. For an electron in the conduction band this is equal to its kinetic energy so that

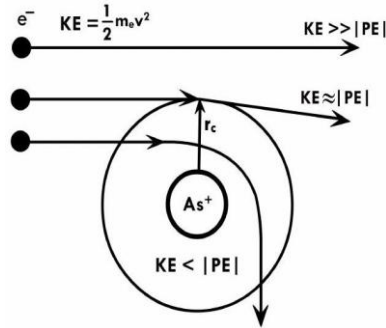
$$\frac{1}{2} m_e^* v_{th}^2 = \frac{3}{2} k_B T \Rightarrow v_{th} \propto \sqrt{T}$$

It can be shown that: $\mu \propto T^{-3/2}$ (lattice scattering)

Since the scattering time is directly proportional to the mobility, this means that μ is also proportional to $T^{(-\frac{3}{2})}$. Thus, the effect of lattice scattering is to reduce the mobility with increase in temperature. This can be understood qualitatively by seeing that as the temperature increases the atomic vibrations increase and at the same time the electrons also travel faster. Both these effects combine to reduce the mobility. Despite this decrease in mobility, conductivity increases with temperature because carrier concentration increases exponentially with temperature.

Impurity scattering:

Impurity scattering becomes important for doped semiconductors. At room temperature the impurities are ionized, so that there is an electrostatic attraction between the electrons traveling in the lattice and the impurity, as shown in the figure for As doped in Si (donor). To find the scattering cross section due to the ionized impurities we can equate the kinetic energy (KE) of the electrons to the potential energy (PE) due to the ionized impurity.



The electron will not be scattered if KE dominates while scattering occurs if PE dominates. The scattering radius, r_c , is the distance at which both effects are equal

$$\frac{3}{2} k_B T = \frac{e^2}{4\pi\epsilon_0\epsilon_r r_c}$$

$$S = \pi r_c^2 \propto T^{-2}$$

So the scattering time due to impurities, τ_I , is given by :

$$\tau_I = \frac{1}{(\pi r_c^2) v_{th} N_I} \propto \frac{1}{(T^{-2}) \left(T^{\frac{1}{2}}\right)} \propto T^{\frac{3}{2}}$$

Again, since the scattering time is directly related to mobility, the mobility due to impurities increases with temperature, opposite to the behavior observed with the lattice. This can be understood qualitatively by considering that with increasing temperature, electrons can travel faster and this makes it easier to escape the ionized impurities.

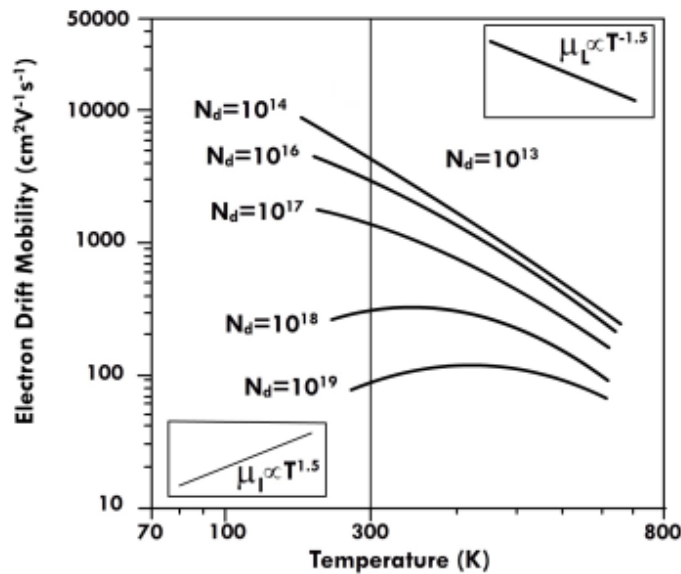
If the impurities are n (Nd-Na) : $\mu \propto T^{\frac{3}{2}}/n$ (doping effect)

Total effect on mobility:

In intrinsic semiconductors the only factor that affects mobility is the lattice scattering so that mobility always reduces with temperature. In extrinsic semiconductors there are two contributions - lattice (μ_L) and the impurities (μ_I). So the total mobility is given by Matthiessen's rule, which is

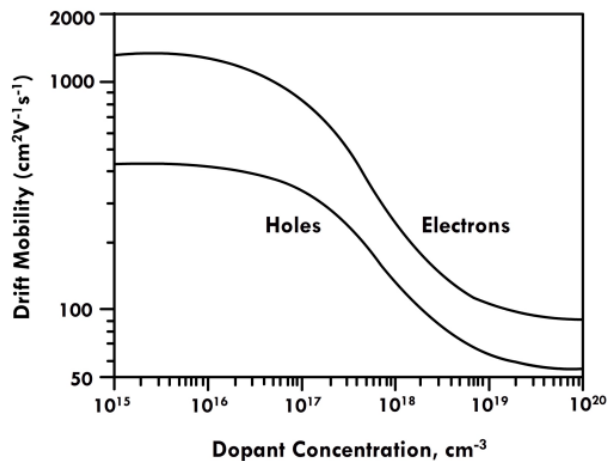
$$\frac{1}{\mu_e} = \frac{1}{\mu_I} + \frac{1}{\mu_L}$$

The scattering process with the lowest mobility dominates. Also, at low temperatures mobility actually increases with temperature since impurity scattering dominates, while at high temperature lattice scattering dominates. This is shown in the figure .



At low dopant concentrations lattice scattering dominates over the entire temperature range. With increasing dopant concentrations impurity scattering starts to dominate at low temperature. A similar behavior is also seen for hole mobility.

Also, at constant temperature the mobility decreases with increasing dopant concentration since there is greater concentration of impurities, as shown in the figure below .



While there is considerable scatter in the experimental data, for noncompensated material (no counter doping), for heavily doped substrates (i.e. 10^{18}cm^{-3} and up), the mobility in silicon is often characterized by the empirical relationship:

$$\mu = \mu_o + \frac{\mu_1}{1 + \left(\frac{N}{N_{\text{ref}}}\right)^\alpha}$$

where N is the doping concentration (either N_D or N_A), and N_{ref} and α are fitting parameters. At room temperature, the above equation becomes:

Majority carriers:

$$\begin{aligned}\mu_n(N_D) &= 65 + \frac{1265}{1 + \left(\frac{N_D}{8.5 \times 10^{16}}\right)^{0.72}} \\ \mu_p(N_A) &= 48 + \frac{447}{1 + \left(\frac{N_A}{6.3 \times 10^{16}}\right)^{0.76}}\end{aligned}$$

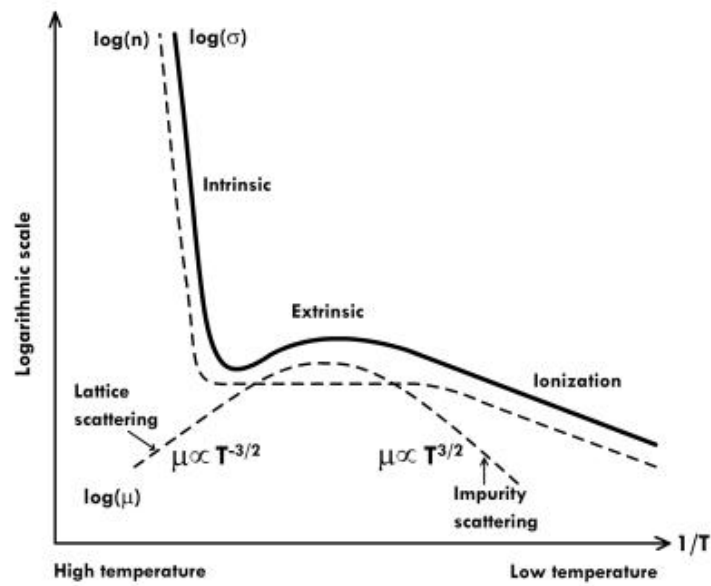
Minority carriers:

$$\begin{aligned}\mu_n(N_A) &= 232 + \frac{1180}{1 + \left(\frac{N_A}{8 \times 10^{16}}\right)^{0.9}} \\ \mu_p(N_D) &= 130 + \frac{370}{1 + \left(\frac{N_D}{8 \times 10^{17}}\right)^{1.25}}\end{aligned}$$

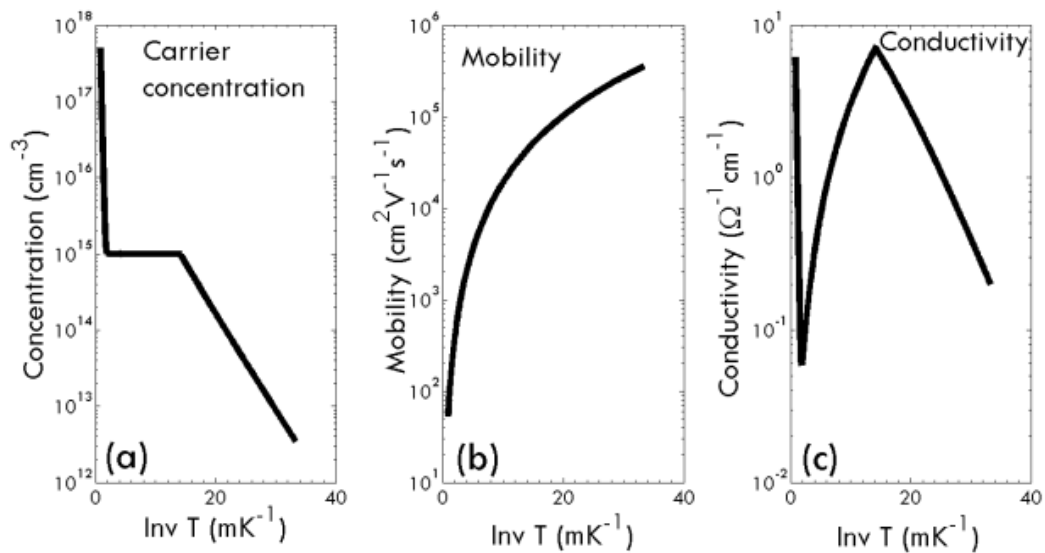
These equations apply only to silicon, and only under low field.

Looking at the temperature dependence of conductivity, both the carrier concentration change and mobility change with temperature and influence the conductivity. The temperature dependence of conductivity is shown in the figure below.

The dominating term is still the carrier concentration, due to its exponential dependence on T but the mobility term also plays a role, especially in the extrinsic regime (or saturation regime). Again, based on conductivity there is an optimal temperature range where σ is nearly constant. For Si, this is around room temperature so that **doping not only gives a temperature independent carrier concentration but also conductivity.**



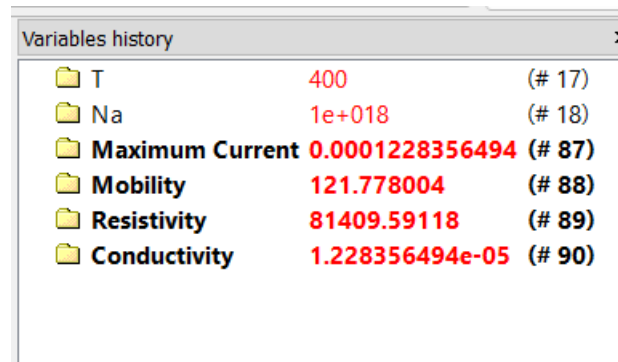
For Si doped with 10^{15} cm^{-3} donors, the carrier concentration, mobility, and conductivity are plotted in the figure below.



Simulation and results :

-We will always use a fixed temperature at 400 K , because we are interested in doping effect on the electrical characteristics of Si.

Result for individual run at $N_a=10^{18} \text{ cm}^{-3}$:



Variables history		
T	400	(# 17)
Na	1e+018	(# 18)
Maximum Current	0.0001228356494	(# 87)
Mobility	121.778004	(# 88)
Resistivity	81409.59118	(# 89)
Conductivity	1.228356494e-05	(# 90)

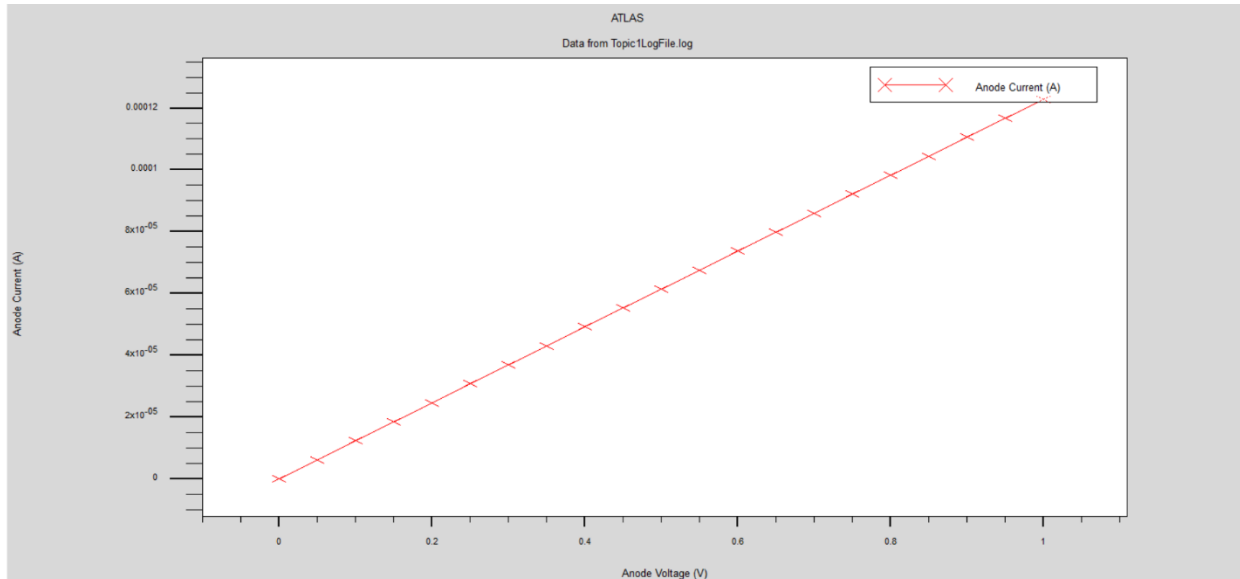
-At a high acceptor doping concentration of $N_a = 10^{18} \text{ cm}^{-3}$ Silicon becomes strongly p-type, leading to a significant reduction in carrier mobility due to enhanced impurity scattering. The mobility decreases to approximately $121.78 \text{ cm}^2/\text{V}\cdot\text{s}$, which is much lower than in intrinsic or lightly doped silicon. ($200\text{-}300 \text{ cm}^2/\text{V}\cdot\text{s}$ for the intrinsic one at 400K).

-Conductivity is also higher (12.2 S/cm) than the intrinsic Si (8.34×10^{-4}) at the same temperature, because the concentration of carriers is much higher (mobility is lower, but doping concentration has the dominant effect)

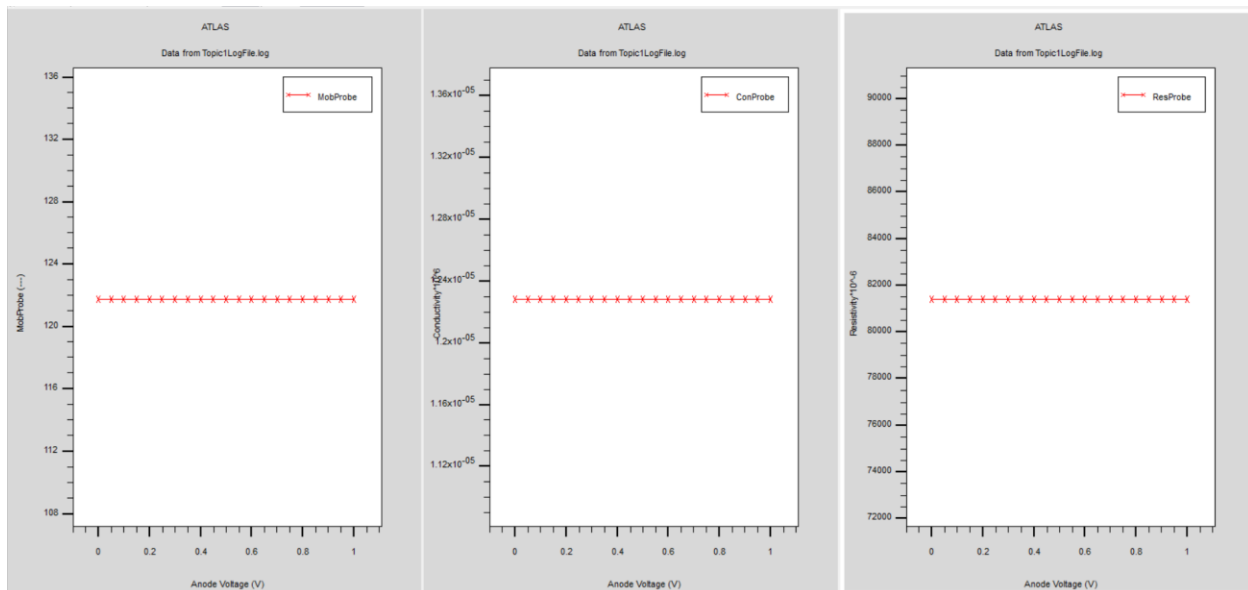
-Since conductivity is higher, we expect a lower resistivity since it is the reciprocal of it .

-There is a voltage applied on the $y=0$ and $y=10$ micrometers. The anode potential is changing from 0 to 1, and the cathode is grounded. So, the Maximum current is when the anode potential=1 V and $I(\text{max})=122.84 \text{ mA}$

I-V characteristics for Si under constant temperature and doping concentration (Temp=400K , Na=1e18 cm⁻³):



Which is a linear relationship as expected.

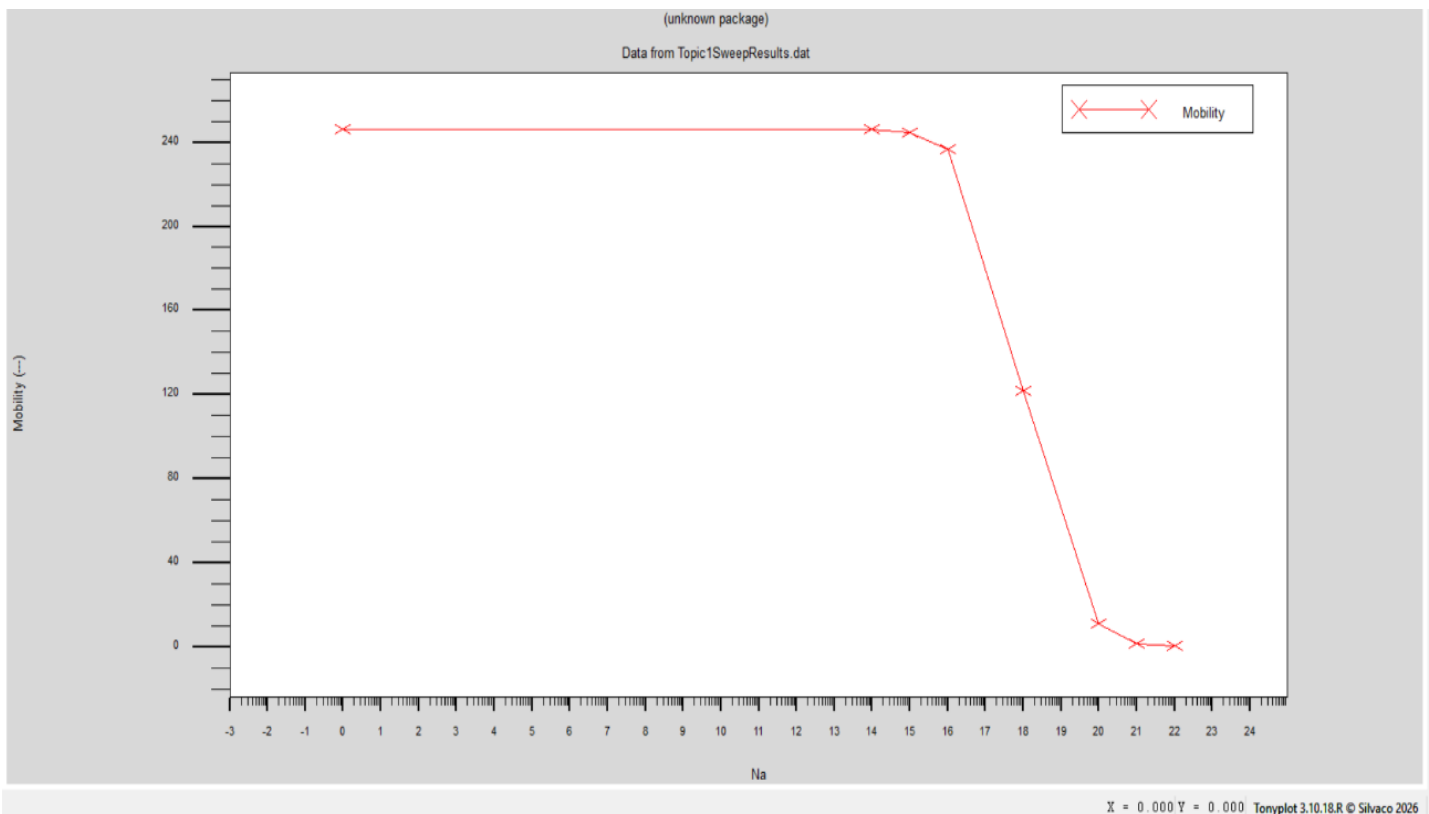


All properties do not change with the applied voltage as expected (not high electric field).

Effect of changing doping concentration on the electrical Characteristics of Si at 400K:

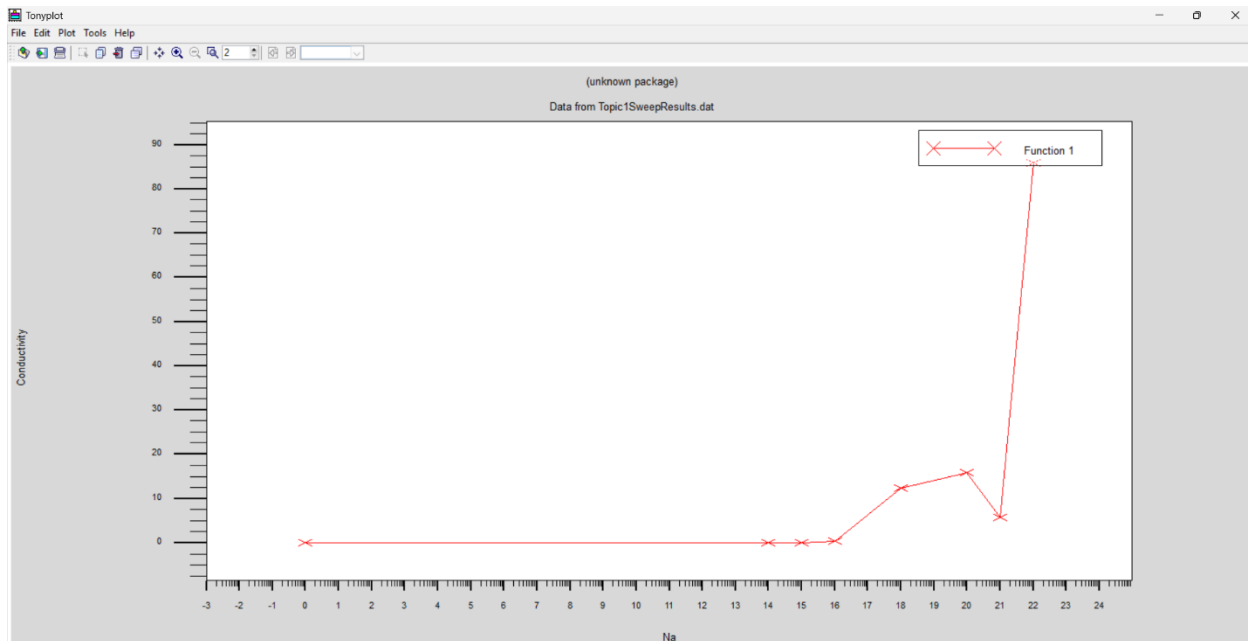
-Now we will sweep the doping concentration N_a ("0, 1e14, 1e15, 1e16, 1e18, 1e20, 1e21, 1e22") under constant temperature 400K .

Effect on Mobility:



The simulated data follows the expected empirical mobility models. The sharp decline in mobility starting around $N_a = 10^{16}$ cm $^{-3}$ highlights the impact of ionized impurities acting as scattering centers, which reduce the mean free path of the carriers. This inverse relationship is critical for device design, as it demonstrates that while higher doping increases carrier concentration, it simultaneously penalizes the carrier velocity.

Conductivity:



The electrical conductivity exhibits a direct correlation with the increase in acceptor doping density. While carrier mobility degrades at high doping levels due to ionized impurity scattering, the exponential increase in hole concentration (p) remains the dominant factor, resulting in a net increase in conductivity. The material transitions from a semi-insulating state to a highly conductive state as N_a exceeds 10^{17} cm^{-3} . The observed fluctuations at extreme doping limits 10^{21} cm^{-3} reflect the critical regime where mobility degradation nearly offsets the gains in carrier density.

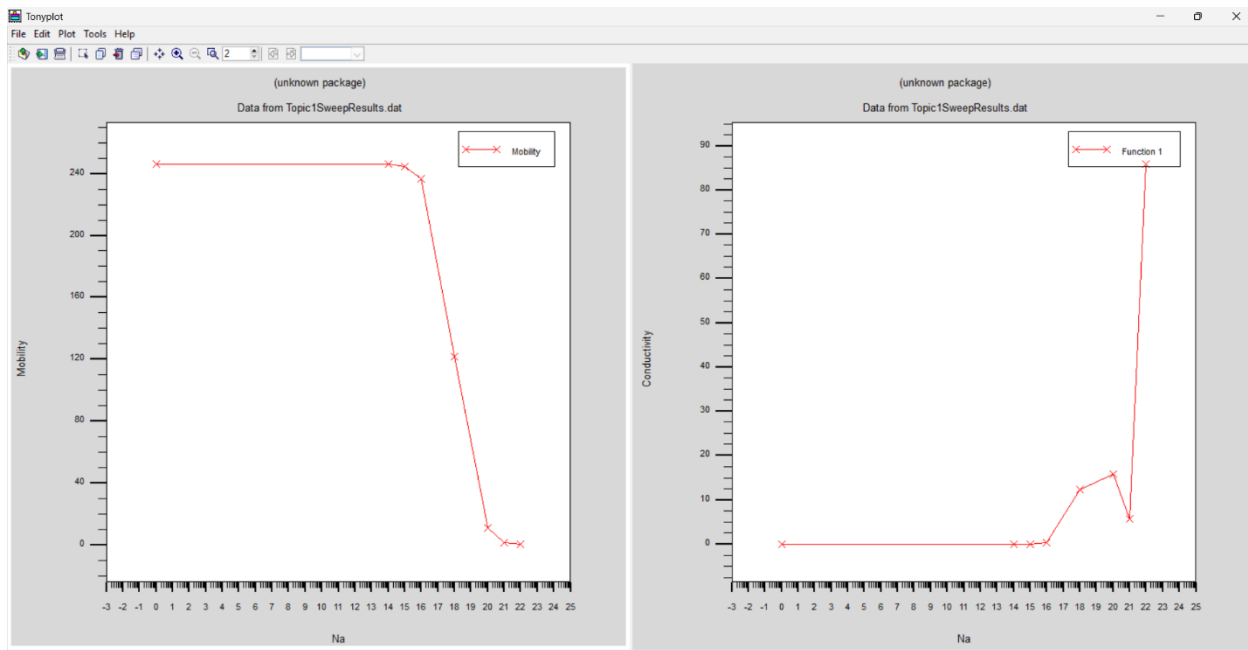
Analysis of Doping-Dependent Conductivity

"The relationship between conductivity and acceptor concentration was evaluated to determine the transport limits of the device. the conductivity is a product of competing physical mechanisms (mobility and doping concentration).

Low-Doping Regime $N_a < 10^{16} \text{ cm}^{-3}$: Conductivity remains negligible as the low carrier density is insufficient for significant charge transport, despite peak mobility values.

High-Doping Regime $N_a > 10^{17} \text{ cm}^{-3}$: A rapid non-linear rise in conductivity is observed. In this region, the increase in the number of available charge carriers mathematically outweighs the reduction in hole mobility caused by impurity scattering.

Degenerate Limit: At concentrations approaching 10^{22} cm^{-3} , the conductivity reaches its maximum, marking the limit of the material's current-carrying capacity under the simulated mobility model.



-Mobility and Conductivity versus N_a (function 1 is conductivity).

The figures illustrate the fundamental trade-off between carrier concentration and transport efficiency in a p-type semiconductor:

-Mobility : Remains stable at low doping but degrades sharply beyond $N_a = 10^{16} \text{ cm}^{-3}$ due to increased ionized impurity scattering.

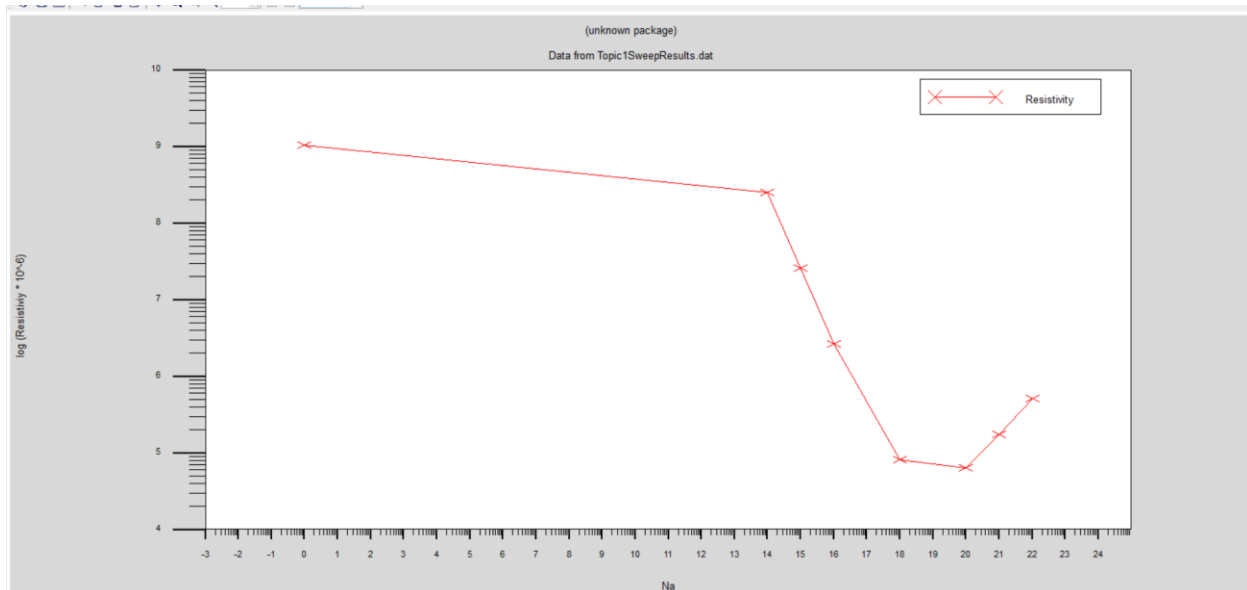
-Conductivity: Despite the drop in mobility, total conductivity rises significantly as N_a exceeds 10^{17} cm^{-3} . This indicates that the

exponential increase in hole concentration mathematically outweighs the reduction in mobility.

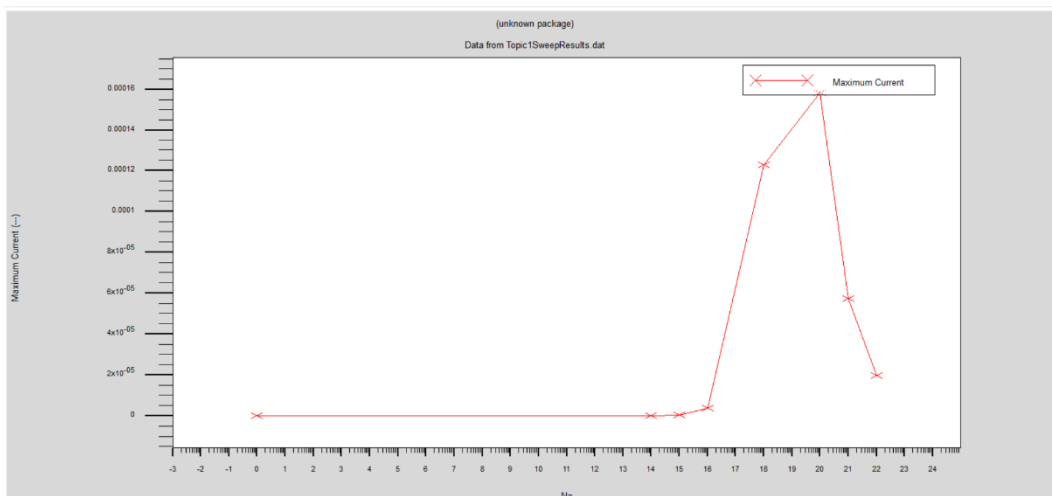
- The material transitions from semi-insulating to highly conductive at high doping levels, with the maximum conductivity achieved at $N_a = 10^{22} \text{ cm}^{-3}$

Resistivity:

of course, it will be the exact opposite behavior of conductivity, because it is the reciprocal of it.



Maximum current in for every N_a relation :



The device's maximum current capacity remains negligible for acceptor concentrations below 10^{16} cm^{-3} , primarily due to the low intrinsic carrier density. A significant peak in current is observed at $N_a = 10^{20} \text{ cm}^{-3}$, where the high carrier concentration optimizes the drive current. However, beyond this threshold, the current degrades; this is attributed to the drastic reduction in carrier mobility at extreme doping levels, which eventually offsets the gains in carrier density.

Conclusion:

The analysis of the simulated data characterizes the intrinsic relationship between doping concentration and the resulting transport properties of the semiconductor. The results confirm that as the acceptor concentration N_a increases, carrier mobility undergoes a significant reduction. This trend aligns with established theoretical models of ionized impurity scattering, where a higher density of dopant atoms increases the frequency of carrier collisions, thereby reducing the mean free path.

Simultaneously, the study of conductivity and maximum current reveals a critical performance threshold. While the mobility of individual carriers decreases, the total electrical conductivity rises sharply due to the exponential increase in the majority carrier population. This confirms the theoretical expectation that in the high-doping regime, the gain in carrier density is the dominant factor governing the material's current-carrying capacity.

The convergence of these trends identifies an optimal doping window for device performance. Beyond the degenerate doping limit, the simulation successfully captures the onset of performance saturation and subsequent degradation, where severe mobility loss begins to offset any further increase in carrier concentration. These findings provide a robust validation of the simulation model against theoretical semiconductor physics and offer a clear guideline for optimizing the electrical characteristics of the device structure.